

Ecological evaluation of a marine protected area network: a progressive-change BACIPS approach

L. THIAULT ^{1,2,3,4,†}, L. KERNALÉGUEN,⁵ C. W. OSENBURG ⁶, T. LISON DE LOMA,^{1,2} Y. CHANCERELLE,^{1,2}
G. SIU,^{1,2} AND J. CLAUDET ^{1,2}

¹National Center for Scientific Research, PSL Université Paris, CRILOBE, USR 3278 CNRS-EPHE-UPVD, Maison des Océans,
195 rue Saint-Jacques, 75005, Paris, France

²Laboratoire d'Excellence CORAIL, 98729 Moorea, French Polynesia

³Museum National d'Histoire Naturelle, PALOC, UMR 208 MNHN-IRD, 75231, Paris, France

⁴Center of Applied Ecology and Sustainability (CAPES) and Centro de Conservación Marina, Facultad de Ciencias Biológicas, Pontificia
Universidad Católica de Chile, Alameda 340, Santiago, Chile

⁵School of Life and Environmental Sciences (Burwood Campus), Deakin University, 3220 Geelong, Victoria, Australia

⁶Odum School of Ecology, University of Georgia, 140 E. Green St., Athens, Georgia 30602 USA

Citation: Thiault, L., L. Kernaléguen, C. W. Osenberg, T. Lison de Loma, Y. Chancerelle, G. Siu, and J. Claudet. 2019. Ecological evaluation of a marine protected area network: a progressive-change BACIPS approach. *Ecosphere* 10(2): e02576. 10.1002/ecs2.2576

Abstract. Marine protected area (MPA) networks, with varying degrees of protection and use, can be useful tools to achieve both conservation and fisheries management benefits. Assessing whether MPA networks meet their objectives requires data from Before the establishment of the network to better discern natural spatiotemporal variation and preexisting differences from the response to protection. Here, we use a Progressive-Change BACIPS approach to assess the ecological effects of a network of five fully and three moderately protected MPAs on fish communities in two coral reef habitats (lagoon and fore reef) based on a time series of data collected five times (over three years) Before and 12 times (over nine years) After the network's establishment on the island of Moorea, French Polynesia. At the network scale, on the fore reef, density and biomass of harvested fishes increased by 19.3% and 24.8%, respectively, in protected areas relative to control fished areas. Fully protected areas provided greater ecological benefits than moderately protected areas. In the lagoon, density and biomass of harvested fishes increased, but only the 31% increase in biomass in fully protected MPAs was significant. Non-harvested fishes did not respond to protection in any of the habitats. We propose that these responses to protection were small, relative to other MPA assessments, due to limited compliance and weak surveillance, although other factors such as the occurrence of a crown-of-thorns starfish outbreak and a cyclone after the network was established may also have impeded the ability of the network to provide benefits. Our results highlight the importance of using fully protected MPAs over moderately protected MPAs to achieve conservation objectives, even in complex social-ecological settings, but also stress the need to monitor effects and adapt management based on ongoing assessments.

Key words: conservation; coral reef; impact assessment design; marine spatial planning; partially protected areas; progressive-change BACIPS; protection regime.

Received 9 December 2018; accepted 17 December 2018. Corresponding Editor: Debra P. C. Peters.

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† **E-mail:** lauric.thiault@gmail.com

INTRODUCTION

Marine protected areas (MPAs) are an important management tool to conserve or restore fish

populations inside their borders (Kerwath et al. 2013, Lubchenco and Grorud-Colvert 2015) and export biomass to surrounding fishing grounds (McClanahan and Mangi 2000, Goñi et al. 2008,

Harmelin-Vivien et al. 2008, Di Lorenzo et al. 2016). Studies have shown that ecological effects of MPAs depend on MPA age (Claudet et al. 2008, Molloy et al. 2009, Friedlander et al. 2017), network design (Jupiter and Egli 2011, Green et al. 2014, 2015), species traits (McClanahan et al. 2007, Claudet et al. 2010, Abesamis et al. 2014), and degree of compliance (Guidetti et al. 2008, Campbell et al. 2012, Gill et al. 2017). The most compelling evidence for beneficial effects of MPAs arises from meta-analyses that synthesize data from many empirical studies (Cote 2001, Micheli et al. 2004, Claudet et al. 2008, Lester et al. 2009, Gill et al. 2017).

Despite accumulated evidence suggesting far-reaching benefits of MPAs, assessment designs of individual MPAs or MPA networks often have substantive limitations, including the lack of data from Before the establishment of the MPA (Willis et al. 2003). As a result, most assessments cannot discern effects of the MPA from preexisting differences or spatiotemporal variability (Guidetti 2002, Halpern et al. 2004, Osenberg et al. 2006, 2011). Even studies with Before samples tend to have a limited time series, often only one survey, making it difficult to attribute any observed temporal trends to effects of the MPA or MPA network (Osenberg et al. 2006, 2011).

When Before data are available, the BACIPS (Before–After Control–Impact Paired-Series) assessment design (Stewart-Oaten et al. 1986, Osenberg et al. 1994, Stewart-Oaten and Bence 2001) provides a powerful tool to overcome many of the limitations of typical studies (e.g., Guidetti 2002, Osenberg et al. 2006, 2011). Repeated assessments Before enforcement provide an estimate of the spatial variability between the Control and Impact sites in the absence of an effect of the MPA. In its simplest application, a change from Before to After the establishment of the MPA in the differences (Δ) in density (or other response parameter) between the Control and Impact sites (i.e., $\Delta_{\text{After}} - \Delta_{\text{Before}}$) provides an estimate of the local effect of the MPA (see Stewart-Oaten and Bence 2001 for a more detailed discussion of the BACIPS analysis and Osenberg et al. 2006, 2011 for discussion of the BACIPS method applied to marine reserves). However, this step-change from Δ_{Before} to Δ_{After} is unlikely in most MPA systems. For example, enforcement after MPA establishment may be gradual or the response of long-lived species may

be slow to accumulate (e.g., Russ and Alcala 2010). In such cases, the effect of the MPA may follow more complex dynamics (Babcock et al. 2010). Recently, a more flexible approach, the Progressive-Change BACIPS, was proposed (Thiault et al. 2017b) that allows quantification of various patterns of temporal change (e.g., linear, asymptotic, sigmoid) in addition to the traditional step-change. Although the need for Before data is widely acknowledged when implementing and assessing an MPA (Claudet and Guidetti 2010) and although BACIPS studies are increasingly used (Castilla and Bustamante 1989, Claudet et al. 2006, Lincoln-Smith et al. 2006, Shears et al. 2006, Moland et al. 2013, Grorud-Colvert et al. 2014, Fletcher et al. 2015), they remain rare (Osenberg et al. 2011).

Existing MPA assessments have focused almost exclusively on no-take, or fully protected, marine reserves. Partially protected areas, where some extractive activities are allowed but still regulated, are often used to balance tradeoffs between conservation and exploitation. Partially protected MPAs are often favored to reach international targets of MPA coverage over the more arduous task of seeking approval for fully protected areas (Lubchenco and Grorud-Colvert 2015, Claudet 2018). Existing assessments of partially protected areas show that their ecological benefits are smaller than those achieved with full protection (Shears et al. 2006, Lester and Halpern 2008, Di Franco et al. 2009, Giakoumi et al. 2017, Zupan et al. 2018).

Here, we assess the ecological effectiveness of a network of eight MPAs, consisting of five fully and three moderately protected MPAs (*sensu* Horta e Costa et al. 2016) in Moorea, French Polynesia. We use a 12-yr time series with three years of Before data and nine years of After data. We apply the Progressive-Change BACIPS design to detect and quantify the pattern of response of fish communities to the establishment of the MPA network. We assessed the effects of the MPA network as a whole, and at the sub-network scales of fully and moderately protected MPAs.

METHODS

Data collection

The MPA network in Moorea, French Polynesia, was officially designated in October 2004,

although establishment and enforcement required several additional years (Lison de Loma et al. 2008). Marine protected areas were delimited inside the lagoon using buoys in September 2005. The first information campaign and police patrols were conducted in 2006. A second information campaign was initiated in 2007, and police monitoring was subsequently increased (and accomplished by hiring a local mediator/enforcement agent and purchasing a boat). We therefore consider 1 January 2007 to constitute the start of enforcement (Lison de Loma et al. 2008).

Marine protected areas within the network were classified using the Regulation-Based Classification System for MPAs (Horta e Costa et al. 2016). The network consists of five fully protected MPAs where all fishing activities are prohibited (i.e., equivalent to no-take zones) and three moderately protected MPAs where restrictions only apply to particular fishing practices and species (Fig. 1; Appendix S1: Table S1). Other areas are open to fishing but subject to general restrictions, such as species size regulations. Within these other areas, we selected five Control sites that were paired with one, two, or three of the MPA sites (Fig. 1) based upon their geographic proximity and physical characteristics.

Fish communities and benthic assemblages were sampled from 2004 to 2015, with sampling surveys (which we refer to as “dates”) conducted once in 2004 (during the dry season), twice each year from 2005 to 2009 (during both the dry and wet seasons), and once each year thereafter (during the wet season). Thus, our dataset consists of five sets of surveys from the Before period (i.e., prior to January 2007) and 12 sets of surveys from the After period (i.e., after January 2007). We refer to each set of surveys as a date, even though data across the whole island were obtained over an approximately week-long period.

At each MPA and Control site, we sampled one location on the fore reef and two locations in the lagoon (Fig. 1). Fishes were identified to species and enumerated along 3, 25 × 2 m underwater belt transects at each location. Total length of each fish was estimated to the nearest centimeter for isolated fish, and mean length was estimated for schools of fish.

We also quantified the density of crown-of-thorns starfish (COTS, *Acanthaster planci*), a coral predator, as well as the cover of live coral and

algae. Coral cover and algal cover were estimated using a point-intercept transect method, using the line that was deployed for fish transects. A total of 50 points were used, spaced equally along each transect. All point contacts were done after the fish surveys to minimize disruption to the fish assemblage.

The progressive-change BACIPS design and analysis

We categorized fish species as harvested or non-harvested based on local expert knowledge and converted all lengths to wet mass (g) using species-specific length–mass relationships (Kulbicki et al. 2005). Sharks, rays, and pelagic species were excluded from the analyses because the transects were not designed to count those highly vagile species. Within each date and habitat, data were averaged across transects ($n = 3$ transects for each fore reef site and $n = 6$ transects for each lagoon site). For each habitat (lagoon or fore reef) and fish group (harvested or non-harvested), we then determined the difference, Δ , between the MPA and its paired Control site (see Fig. 1) after log-transformation (Lison de Loma et al. 2008):

$$\Delta_{P,i} = \ln(N_{\text{MPA},P,i} + a) - \ln(N_{\text{Control},P,i} + a) \quad (1)$$

where N is the average target fish density or biomass (across the three or six transects) at either the MPA or Control site, during the i^{th} date in the P^{th} period ($P = \text{Before}$ or $P = \text{After}$), and a is a constant that was added to avoid taking logarithms of zero. To use the smallest plausible value of “ a ,” we chose a value for each fish group that represented the addition of one fish to one of the n ($n = 3$ or 6) transects (i.e., $a = 1/n$ for analyses of density, and $a = [\text{average mass of a fish in all surveys}]/n$ for analyses of biomass).

We evaluated whether protection had an ecological effect by assessing if the difference in density or biomass, Δ , changed from Before to After the establishment of the MPAs. The classic BACIPS approach assumes a step-change in Δ . Instead of a step-change, we hypothesized that the temporal change in Δ might exhibit a more complex pattern because (1) fish assemblages can take several decades after protection to fully recover from fishing (Russ and Alcala 2004, Babcock et al. 2010, MacNeil et al. 2015), and, perhaps more importantly, (2) enforcement of the

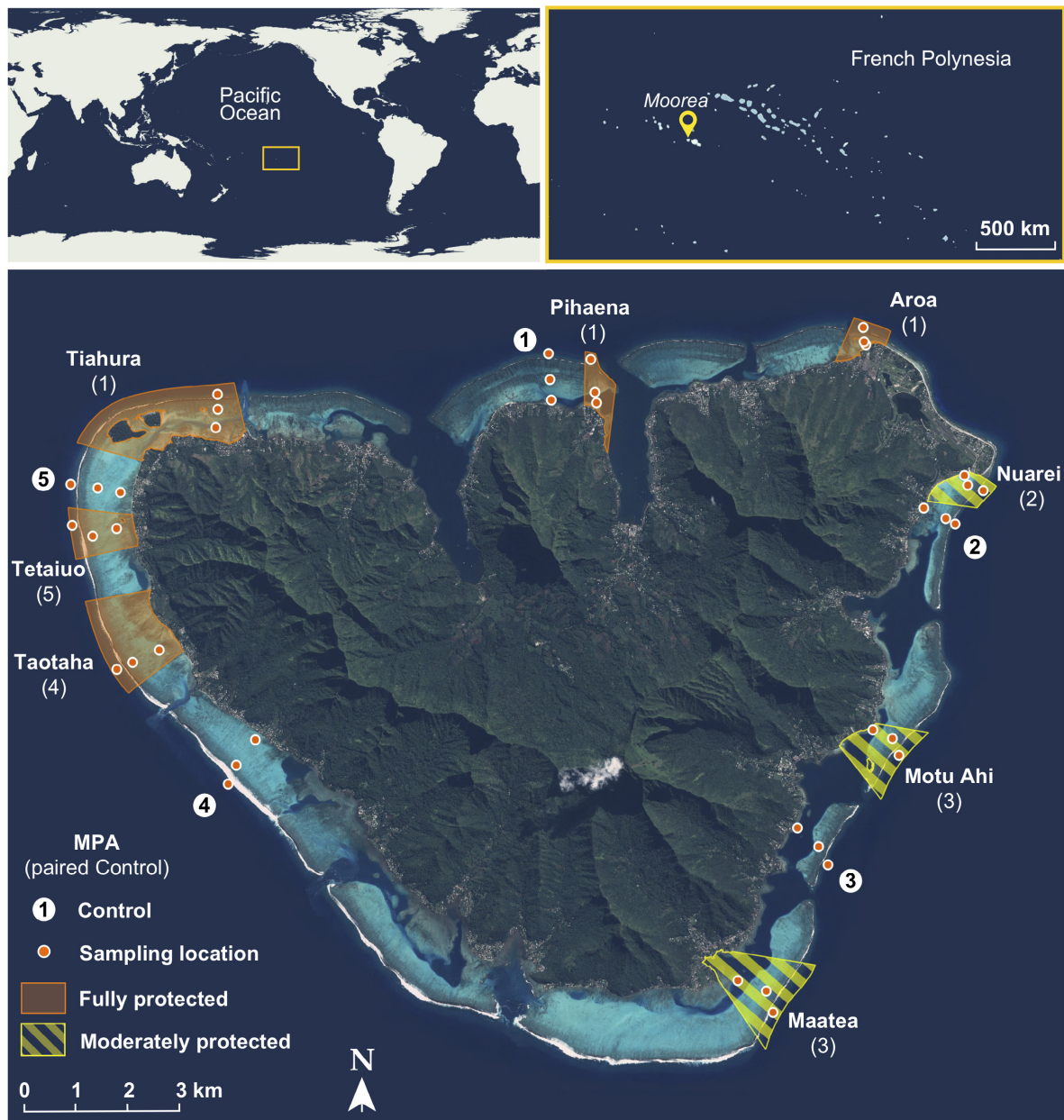


Fig. 1. Network of marine protected areas (MPAs) around the island of Moorea, French Polynesia. The network consists of five fully protected MPAs (on the north and west shores) and three moderately protected MPAs (on the east shore) using the regulation-based classification system for MPAs (Horta e Costa et al. 2016). Numbers in parentheses after MPA names refer to the Control site to which it was paired. Sampling at each MPA and Control was conducted in two distinct habitats (in the lagoon at two locations and on the fore reef at one location), five times (over three years) Before and 12 times (over nine years) After implementation.

MPAs in the network was gradually phased in after January 2007. Therefore, we applied a Progressive-Change BACIPS by comparing four competing models: step-change (Eq. 2), linear

(Eq. 3), asymptotic (Eq. 4), and sigmoid models (Eq. 5). As for any BACIPS analysis, we assumed that in the Before period, Δ was stationary through time (i.e., there was no consistent

temporal trend in Δ_{Before}), but After enforcement, Δ changed with time, with an increase indicating a positive effect of the MPA and a decrease indicating a negative effect. Thus, the differences between an MPA and its Control After enforcement (Δ_{After}) could be described as

$$\text{Step-change response: } \Delta_{\text{After},i} = M + \Delta_{\text{Before}} + \varepsilon_i \quad (2)$$

$$\text{Linear response: } \Delta_{\text{After},i} = rt_i + \Delta_{\text{Before}} + \varepsilon_i \quad (3)$$

$$\text{Asymptotic response: } \Delta_{\text{After},i} = \frac{Mt}{L+t} + \Delta_{\text{Before}} + \varepsilon_i \quad (4)$$

$$\text{Sigmoid response: } \Delta_{\text{After},i} = \frac{M(t/L)^K}{1 + (t/L)^K} + \Delta_{\text{Before}} + \varepsilon_i \quad (5)$$

where t is time (in years) since 1 January 2007 associated with the i^{th} sampling survey; Δ_{Before} is the underlying spatial variation between the two sites in the absence of an MPA effect (and was estimated during the Before period); r and M are the rate of divergence and magnitude of change between the MPA site and Control (due to the establishment of the MPA), respectively; L is the time required to achieve half of the long-term effect; K is a scaling parameter in the sigmoid model that affects the steepness of the curve; and ε is the error associated with the i^{th} survey. Each model was fitted setting all Before data to have $t = 0$, and the magnitude of the response of fish to protection (hereafter referred to as effect size) was then measured based on the predictions of the best-fit model (highest AICc score) at $t = 9$ (i.e., 2015), the last year of the survey (Thiault et al. 2017b).

Having first checked assumptions of normality, homogeneity of variance, and absence of autocorrelation (Appendix S1: Table S2), we assessed the effect of (1) the whole network (i.e., the island-wide effect of the MPA network), (2) the two sub-networks (i.e., the networks of fully protected and moderately protected MPAs), and (3) individual MPAs. To avoid pseudo-replication, and because we expected that each MPA would have its own response, we included MPA as a random effect for all the parameters describing the effect size (i.e., M , r , K , and L). Season was not included

in the model to maximize statistical power and because the BACIPS design reduces the effect of seasonality. All analyses were performed using the R statistical software (R Core Team 2017).

RESULTS

Fishes were generally more abundant on the fore reef compared to the lagoon (Fig. 2). Coral and algal cover changed dramatically over the period of study on the fore reef following a crown-of-thorns starfish (COTS; *Acanthaster planci*) outbreak between 2007 and 2012 and Cyclone Oli in 2010: As COTS increased, coral cover decreased by more than 90% between October 2006 and March 2010, and algae increased by 34% (Fig. 2).

Effect of protection on harvested and non-harvested fishes

Estimated statistical power to detect MPA effects was high at the MPA network and sub-networks scales, suggesting that we would be able to detect changes if they did occur. For example, the probability of detecting a 100% increase (typically seen in other MPA assessments) was 0.97 on average across the 24 comparisons (Appendix S1: Fig. S2). Power to detect responses at individual MPAs was considerably lower. Therefore, we focus on the network and sub-network responses; results for individual MPAs are presented in Appendix S1 (Table S3).

The pattern of change in response to protection provided more support for the step-change model (asserting an immediate shift in density in the MPA relative to the Control site) than for the linear model (in which the difference increases linearly with time since protection). In no case was the asymptotic or sigmoid model better supported by the data. In 25% of cases, the likelihood (ω) of the second best-fit model was comparable to that of the best-fit model (i.e., $\omega_{\text{best-fit}} - \omega_{\text{second best-fit}} \leq 10\%$). For consistency, we only present results derived from the best-fit model (Fig. 3).

On the fore reef, at the network scale, harvested fish biomass increased significantly by 24.8% in MPAs relative to the paired Controls (Fig. 3); the relative increase in density was similar in magnitude (19.3%), although not significant. This increase was mostly due to the response in fully protected MPAs (where all

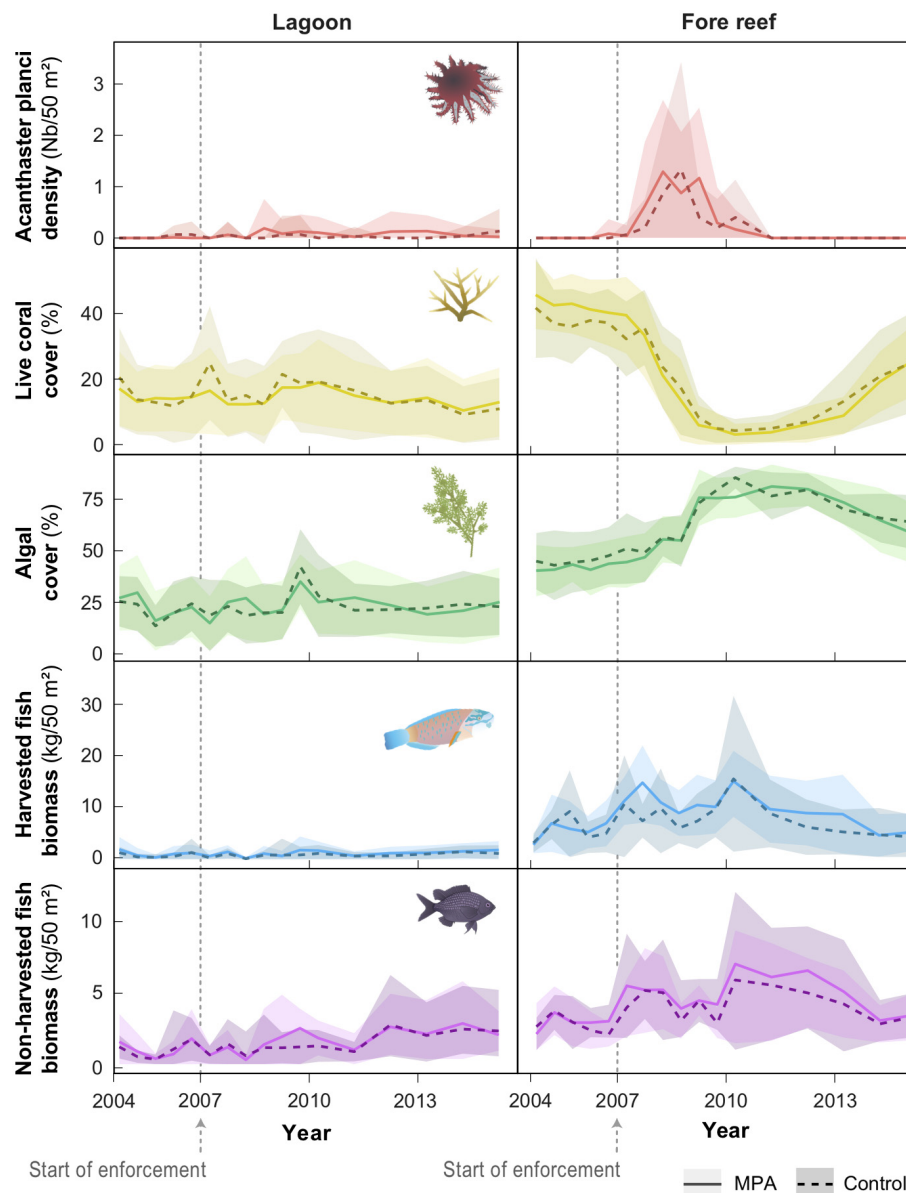


Fig. 2. Temporal variation in key ecological components of the studied coral reef ecosystem. From top to bottom: density of *Acanthaster planci* (red), living coral cover (yellow), algal cover (green), biomass of harvested (blue), and non-harvested fish (purple) in the lagoon (left column) and fore reef (right column). Lines are the mean of the eight marine protected areas (solid line) and five Control sites (dashed line), and ribbons represent standard deviations (the darker ribbon corresponds to the Controls). Marine protected area enforcement started in January 2007, which separates the Before (2004–2006) from the After (2007–2015) periods.

fishing is prohibited), where density and biomass of harvested fish increased significantly by 43.2% and 31%, respectively. No significant increase in either density or biomass was observed for the sub-network of moderately protected MPAs,

where fishing restrictions apply to only a subset of fishing methods and species (Fig. 3). Protection benefited herbivores, invertivores, and piscivores (Appendix S1: Fig. S2), the large majority of which are targeted by fishing.

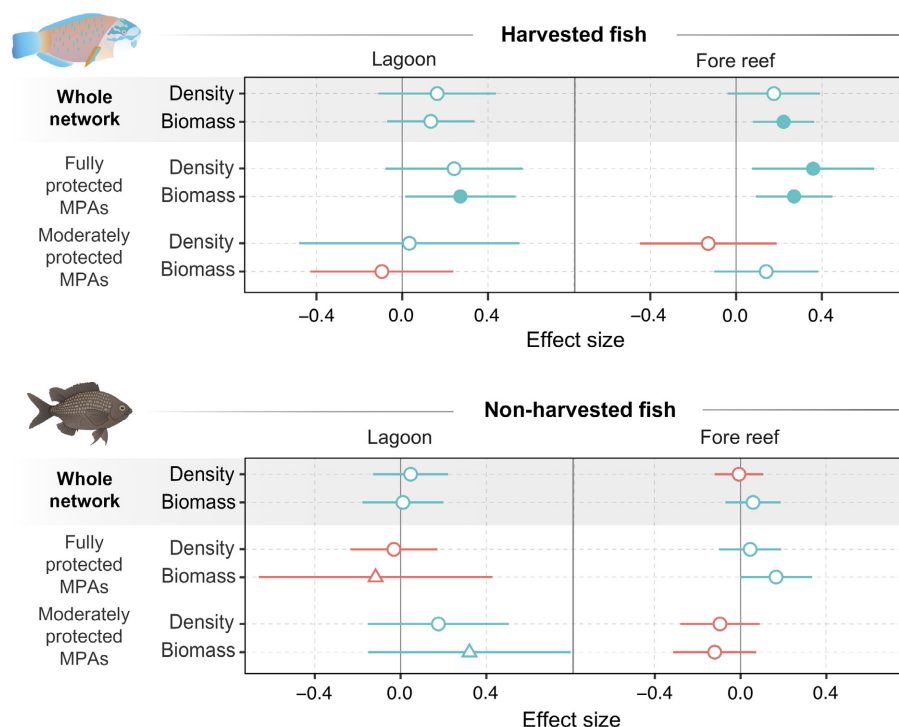


Fig. 3. Effect sizes measured at the whole network scale, and for fully protected marine protected areas (MPAs) and moderately protected MPAs separately, in the lagoon and fore reef, for density and biomass of harvested and non-harvested fishes. Effect sizes are expressed as the change in the log-ratio of the density or biomass in the MPA relative to its Control (Δ) from $t = 0$ (i.e., <2007) to $t = 9$ (i.e., 2015) as predicted by the best-fit model. Changes by a factor of 2 correspond to effects sizes of -0.7 (halving) or 0.7 (doubling) over the 9-yr period of protection. Positive effects are depicted in red and negative effects in blue. Filled symbols indicate that the 95% confidence interval of the effect does not overlap zero. Shapes indicate the best-fit model: step-change (circle) and linear (triangle). There were no cases in which the asymptotic or sigmoid model was best supported by the data. See Appendix S1: Table S3 for results at the individual MPA-scale and Appendix S1: Fig. S1 for results broken down by trophic group.

For harvested fish inside the lagoon, the only significant effect that we detected was on biomass in the fully protected MPA sub-network, which showed an increase of 31.1%. Although positive, the effect on densities was non-significant.

Effects on non-harvested fishes were generally smaller in magnitude and more often negative (Fig. 3; Appendix S1: Fig. S2). However, none of these responses differed from zero, whether for the whole network or for the two sub-networks.

DISCUSSION

Disentangling the effects of management interventions from other sources of spatiotemporal variation is a challenging but necessary task to

derive robust inferences that can inform evidence-based decisions. Before-After Control-Impact Paired-Series designs are particularly powerful tools to address this challenge, yet, although there have been hundreds of assessments of MPAs, very few include data from Before the establishment of the MPA and even fewer have multiple surveys from Before (Castilla and Bustamante 1989, Lincoln-Smith et al. 2006, Shears et al. 2006). As a result, most assessments are unable to unequivocally isolate the effects of the MPA from preexisting differences (Osenberg et al. 2006). To our knowledge, this is the first time a Progressive-Change BACIPS approach—which expands the scope of BACIPS analyses beyond the step-change response

assumed in previous assessments—has been used to assess the impact of a network of MPAs on fish communities.

We expected continuous changes in the response of fish to protection due to ecological processes combined with gradual enforcement in the After period. Yet, asymptotic and sigmoid models were never selected among the competing models, and the linear model was only selected twice. The lack of support for complex, non-linear responses likely resulted from the low effect sizes combined with relatively high spatiotemporal variability. This likely led to the selection of simple (step-change and linear) models over more complex ones (Thiault et al. 2017b).

The establishment of the MPA network in Moorea provided positive, but limited, ecological benefits. Network-wide, harvested fish biomass had increased by 24.8% on the fore reef eight years after the start of enforcement, in protected areas relative to Controls. Importantly, this positive, network-scale effect was mostly driven by responses where fishing was entirely prohibited (i.e., in the fully protected MPAs, biomass and density increased +30% and +43.2%, respectively). Indeed, no positive effect was detected in either habitat within moderately protected MPAs (*sensu* Horta e Costa et al. 2016). This result is consistent with a recent meta-analysis that found that fully and highly protected areas consistently conferred ecological benefits, but moderately protected MPAs provided ecological benefits only when immediately adjacent to fully protected areas (Zupan et al. 2018).

The response in fully protected MPAs, although the largest effect we detected, was relatively small compared to effects documented in other published MPA meta-analyses, in which harvested organisms were generally 2–3 times more abundant inside MPAs compared to fishing grounds (Halpern 2003, Claudet et al. 2008, 2011). The lower effect measured in this study could be due to a combination of factors, which we elaborate upon below, including (1) overestimation of the effect size in previous studies due to the lack of Before data (Osenberg et al. 2006, 2011), (2) limited compliance and enforcement (Guidetti et al. 2008, Campbell et al. 2012), and (3) dramatic ecosystem disturbance following enforcement due to the crown-of-thorns starfish (COTS) outbreak and cyclone (Adam et al. 2011, Lamy et al. 2015, 2016).

Marine protected areas are often strategically implemented in sites with higher densities than surrounding areas. The absence of Before data precludes the incorporation of these initial, and potentially large, differences, which may then become confounded with effects of MPAs. As a result, studies that lack Before data may overestimate the benefits of MPAs. Indeed, it has been suggested that up to half of the commonly observed increase in density inside MPAs may be due to these preexisting differences (Osenberg et al. 2006, 2011). Our smaller-than-expected effect sizes support this interpretation. They also support Osenberg et al.'s (2011) conjecture that by identifying Control sites Before MPA enforcement, siting and MPA effects are less likely to be confounded: For example, effect sizes from a Control-Impact analysis of our data yielded effects that were comparable in magnitude to (i.e., not larger than) effects from the BACIPS analyses (Appendix S1: Fig. S3). The Control-Impact comparison, however, revealed two important differences. First, variation in the effects was greater for the Control-Impact analyses, presumably because initial differences between sites added to the variation in the Control-Impact effects (relative to the effects quantified with BACIPS). Secondly, the confidence intervals on the effects were smaller for the Control-Impact estimates, likely because Control-Impact studies only capture spatial variation while BACIPS analyses capture spatiotemporal variation. These results not only suggest that Control-Impact studies might overestimate effects, but that they might also give a false sense of confidence in the estimates because they fail to incorporate spatiotemporal variance. The BACIPS approach circumvents these problems and likely leads to smaller, more accurate, but less precise, estimates of effect sizes.

Limited public appreciation about the benefits of MPAs and an understaffed management team may have limited compliance and therefore ecological outcomes associated with the Moorea MPA network (Gaspar and Bambridge 2008, Guidetti et al. 2008, Edgar et al. 2014, Gill et al. 2017). Surveillance reports made by the local mediator (Gaspar and Bambridge 2008) and surveys completed by local experts (Appendix S1: Table S5) suggest that enforcement was heterogeneous, being mostly limited to the north shore, where the largest beneficial effects were observed

(Appendix S1: Figs. S4 and S5). Furthermore, the lagoon habitat is highly fished at night with light attractors, and surveillance is non-existent at night. The fore reef is less accessible and more hazardous, and, as a result, less likely to experience night poaching (Thiault et al. 2017a), potentially explaining why the responses on the fore reef were greater than in the lagoon (Fig. 3).

The whole island underwent severe natural disturbances since the network's establishment. A COTS outbreak occurred starting in 2007 and was followed by a cyclone in 2010. This resulted in a dramatic (90%) decline in live coral cover, especially on the fore reef, and an increase in turf and macro-algae (Fig. 2). These shifts in micro-habitats led to changes in the composition of reef-associated fishes, but after a time lag of several years (Adam et al. 2011, Lamy et al. 2015, Han et al. 2016). These types of dramatic temporal changes can cause problems with some types of assessments (e.g., Before–After comparisons). Before–After Control–Impact Paired–Series, in theory, can handle such regional phenomena because the Control site should reflect the effects of the regional processes (i.e., the natural disturbance) but not the local factors (i.e., MPA). Surveys confirmed that similar impacts of COTS and the hurricane were simultaneously observed at the MPA and Control sites (Kayal et al. 2012, Lamy et al. 2015). This, combined with the fact that functional groups that are highly sensitive to habitat changes did not respond to protection (Appendix S1: Fig. S2), suggests that observed MPA effects are not confounded with COTS and cyclone effects. These issues underscore the importance of implementing BACIPS designs for future MPA assessments.

Our results have been communicated to some local community members and to local administrations to facilitate the ongoing revision of the marine spatial management plan in Moorea. The marine spatial management plan of Moorea is currently being revised to better engage local communities and foster better compliance. Monitoring of the current and future MPA network in Moorea will continue, but will be revised to increase sampling effort: For example, more numerous and larger transects should facilitate future MPA-specific assessments as well as network-wide results. We believe the Progressive-Change BACIPS approach used here represents a meaningful advance that

could facilitate and inform management interventions in other social–ecological settings.

ACKNOWLEDGMENTS

We thank everyone who contributed to this project over its many years, especially with respect to study design, data collection, and coordination: T. Adam, G. Bergsma, O. Le Bihan, P. Bosserelle, A. Brooks, N. Davies, P. Edmunds, R. Galzin, S. Geange, X. Han, J. Idjadi, A. Jezequel, D. Lecchini, V. Liao, N. Maihota, M. Petit, S. Poujade, N. Price, S. Talmage, J. Shima, A. Stier, C. Vieux, and J. White. Service des Ressources Marines et Minières, Service de l'Environnement, and CRILOBE Service d'Observation CORAIL provided funding for data collection, and we especially thank C. Monier for his early support. Support from ANR (ANR-14-CE03-0001-01), Fondation de France (INTENSE), and NSF (OCE-0242312, OCE-1130359) helped facilitate the study. L. Thiault's PhD grant was funded by Pierre and Marie Curie University (PDI-MSC grant).

LITERATURE CITED

- Abesamis, R. A., A. L. Green, G. R. Russ, and C. R. L. Jadhoc. 2014. The intrinsic vulnerability to fishing of coral reef fishes and their differential recovery in fishery closures. *Reviews in Fish Biology and Fisheries* 24:1033–1063.
- Adam, T. C., R. J. Schmitt, S. J. Holbrook, A. J. Brooks, P. J. Edmunds, R. C. Carpenter, and G. Bernardi. 2011. Herbivory, connectivity, and ecosystem resilience: response of a coral reef to a large-scale perturbation. *PLoS ONE* 6:e23717.
- Babcock, R. C., N. T. Shears, A. C. Alcala, N. S. Barrett, G. J. Edgar, K. D. Lafferty, T. R. McClanahan, and G. R. Russ. 2010. Decadal trends in marine reserves reveal differential rates of change in direct and indirect effects. *Proceedings of the National Academy of Sciences* 107:18256–18261.
- Campbell, S. J., A. S. Hoey, J. Maynard, T. Kartawijaya, J. Cinner, N. A. J. Graham, and A. H. Baird. 2012. Weak compliance undermines the success of no-take zones in a large government-controlled marine protected area. *PLoS ONE* 7:e50074.
- Castilla, J. C., and R. H. Bustamante. 1989. Human exclusion from rocky intertidal of Las Cruces, central Chile: effects on *Durvillaea antarctica* (Phaeophyta, Durvilleales). *Marine Ecology Progress Series* 50:203–214.
- Claudet, J. 2018. Six conditions under which MPAs might not appear effective (when they are). *ICES Journal of Marine Science* 75:1172–1174.
- Claudet, J., J. A. García-Charton, and P. Lenfant. 2011. Combined effects of levels of protection and

- environmental variables at different spatial resolutions on fish assemblages in a marine protected area. *Conservation Biology* 25:105–114.
- Claudet, J., and P. Guidetti. 2010. Improving assessments of marine protected areas. *Aquatic Conservation: Marine and Freshwater Ecosystems* 20: 239–242.
- Claudet, J., D. Pelletier, J.-Y. Jouvenel, F. Bachet, and R. Galzin. 2006. Assessing the effects of marine protected area (MPA) on a reef fish assemblage in a northwestern Mediterranean marine reserve: identifying community-based indicators. *Biological Conservation* 130:349–369.
- Claudet, J., et al. 2008. Marine reserves: Size and age do matter. *Ecology Letters* 11:481–489.
- Claudet, J., et al. 2010. Marine reserves: Fish life history and ecological traits matter. *Ecological Applications* 20:830–839.
- Cote, I. 2001. Effects of marine reserve characteristics on the protection of fish populations: a meta-analysis. *Journal of Fish Biology* 59:178–189.
- Di Franco, A., S. Bussotti, A. Navone, P. Panzalis, and P. Guidetti. 2009. Evaluating effects of total and partial restrictions to fishing on Mediterranean rocky-reef fish assemblages. *Marine Ecology Progress Series* 387:275–285.
- Di Lorenzo, M., J. Claudet, and P. Guidetti. 2016. Spillover from marine protected areas to adjacent fisheries has an ecological and a fishery component. *Journal for Nature Conservation* 32:62–66.
- Edgar, G. J., et al. 2014. Global conservation outcomes depend on marine protected areas with five key features. *Nature* 506:216–220.
- Fletcher, W. J., R. E. Kearney, B. S. Wise, and W. J. Nash. 2015. Large-scale expansion of no-take closures within the Great Barrier Reef has not enhanced fishery production. *Ecological Applications* 25:1187–1196.
- Friedlander, A. M., Y. Golbuu, E. Ballesteros, J. E. Caselle, M. Gouezo, D. Olsudong, and E. Sala. 2017. Size, age, and habitat determine effectiveness of Palau's Marine Protected Areas. *PLoS ONE* 12: e0174787.
- Gaspar, C., and T. Bambridge. 2008. Territorialités et aires marines protégées à Moorea (Polynésie française). *Journal de la Société des Océanistes* 127:231–246.
- Giakoumi, S., et al. 2017. Ecological effects of full and partial protection in the crowded Mediterranean Sea: a regional meta-analysis. *Scientific Reports* 7:8940.
- Gill, D. A., et al. 2017. Capacity shortfalls hinder the performance of marine protected areas globally. *Nature* 543:665–669.
- Goñi, R., et al. 2008. Spillover from six western Mediterranean marine protected areas: evidence from artisanal fisheries. *Marine Ecology Progress Series* 366:159–174.
- Green, A. L., L. Fernandes, G. Almany, R. Abesamis, E. McLeod, P. M. Aliño, A. T. White, R. Salm, J. Tanzer, and R. L. Pressey. 2014. Designing marine reserves for fisheries management, biodiversity conservation, and climate change adaptation. *Coastal Management* 42:143–159.
- Green, A. L., A. P. Maypa, G. R. Almany, K. L. Rhodes, R. Weeks, R. A. Abesamis, M. G. Gleason, P. J. Mumby, and A. T. White. 2015. Larval dispersal and movement patterns of coral reef fishes, and implications for marine reserve network design. *Biological Reviews* 90:1215–1247.
- Gorud-Colvert, K., et al. 2014. Marine protected area networks: assessing whether the whole is greater than the sum of its parts. *PLoS ONE* 9:e102298.
- Guidetti, P. 2002. The importance of experimental design in detecting the effects of protection measures on fish in Mediterranean MPAs. *Aquatic Conservation: Marine and Freshwater Ecosystems* 12:619–634.
- Guidetti, P., et al. 2008. Italian marine reserve effectiveness: Does enforcement matter? *Biological Conservation* 141:699–709.
- Halpern, B. S. 2003. The impact of marine reserves: Do reserves work and does reserve size matter? *Ecological Applications* 13:117–137.
- Halpern, B. S., S. D. Gaines, and R. R. Warner. 2004. Confounding effects of the export of production and the displacement of fishing effort from marine reserves. *Ecological Applications* 14:1248–1256.
- Han, X., T. C. Adam, R. J. Schmitt, A. J. Brooks, and S. J. Holbrook. 2016. Response of herbivore functional groups to sequential perturbations in Moorea, French Polynesia. *Coral Reefs* 35:999–1009.
- Harmelin-Vivien, M., L. Le Diréach, J. Bayle-Sempere, E. Charbonnel, J. A. García-Charton, D. Ody, A. Pérez-Ruzafa, O. Reñones, P. Sánchez-Jerez, and C. Valle. 2008. Gradients of abundance and biomass across reserve boundaries in six Mediterranean marine protected areas: Evidence of fish spillover? *Biological Conservation* 141:1829–1839.
- Horta e Costa, B., J. Claudet, G. Franco, K. Erzini, A. Caro, and E. J. Gonçalves. 2016. A regulation-based classification system for Marine Protected Areas (MPAs). *Marine Policy* 72:192–198.
- Jupiter, S. D., and D. P. Egli. 2011. Ecosystem-based management in Fiji: successes and challenges after five years of implementation. *Journal of Marine Biology* 2011:1–14.

- Kayal, M., et al. 2012. Predator crown-of-thorns starfish (*Acanthaster planci*) outbreak, mass mortality of corals, and cascading effects on reef fish and benthic communities. *PLoS ONE* 7:e47363.
- Kerwath, S. E., H. Winker, A. Götz, and C. G. Attwood. 2013. Marine protected area improves yield without disadvantaging fishers. *Nature Communications* 4:2347.
- Kulbicki, M., N. Guillemot, and M. Amand. 2005. A general approach to length-weight relationships for New Caledonian lagoon fishes. *Cybiu* 29:235–252.
- Lamy, T., R. Galzin, M. Kulbicki, T. Lison de Loma, and J. Claudet. 2016. Three decades of recurrent declines and recoveries in corals belie ongoing change in fish assemblages. *Coral Reefs* 35:293–302.
- Lamy, T., P. Legendre, Y. Chancerelle, G. Siu, and J. Claudet. 2015. Understanding the spatio-temporal response of coral reef fish communities to natural disturbances: insights from beta-diversity decomposition. *PLoS ONE* 10:e0138696.
- Lester, S., and B. Halpern. 2008. Biological responses in marine no-take reserves versus partially protected areas. *Marine Ecology Progress Series* 367:49–56.
- Lester, S., B. Halpern, K. Grorud-Colvert, J. Lubchenco, B. Ruttenberg, S. Gaines, S. Aïramé, and R. Warner. 2009. Biological effects within no-take marine reserves: a global synthesis. *Marine Ecology Progress Series* 384:33–46.
- Lincoln-Smith, M. P., K. A. Pitt, J. D. Bell, and B. D. Madstone. 2006. Using impact assessment methods to determine the effects of a marine reserve on abundances and sizes of valuable tropical invertebrates. *Canadian Journal of Fisheries and Aquatic Sciences* 63:1251–1266.
- Lison de Loma, T., C. W. Osenberg, J. S. Shima, Y. Chancerelle, N. Davies, and A. J. Brooks. 2008. A framework for assessing impacts of marine protected areas in Moorea (French Polynesia). *Pacific Science* 427:427–438.
- Lubchenco, J., and K. Grorud-Colvert. 2015. Making waves: the science and politics of ocean protection. *Science* 350:382–383.
- MacNeil, M. A., et al. 2015. Recovery potential of the world's coral reef fishes. *Nature* 520:341–344.
- McClanahan, T. R., N. A. J. Graham, J. M. Calnan, and M. A. MacNeil. 2007. Toward pristine biomass: reef fish recovery in coral reef marine protected areas in Kenya. *Ecological Applications* 17:1055–1067.
- McClanahan, T. R., and S. Mangi. 2000. Spillover of exploitable fishes from a marine park and its effect on the adjacent fishery. *Ecological Applications* 10:1792–1805.
- Micheli, F., B. S. Halpern, L. W. Botsford, and R. R. Warner. 2004. Trajectories and correlates of community change in no-take marine reserves. *Ecological Applications* 14:1709–1723.
- Moland, E., E. M. Olsen, H. Knutsen, P. Garrigou, S. H. Espeland, A. R. Kleiven, C. Andre, and J. A. Knutsen. 2013. Lobster and cod benefit from small-scale northern marine protected areas: inference from an empirical before-after control-impact study. *Proceedings of the Royal Society B: Biological Sciences* 280:20122679.
- Molloy, P. P., I. B. McLean, and I. M. Côté. 2009. Effects of marine reserve age on fish populations: a global meta-analysis. *Journal of Applied Ecology* 46:743–751.
- Osenberg, C. W., B. M. Bolker, J.-S. S. White, C. M. St. Mary, and J. S. Shima. 2006. Statistical issues and study design in ecological restorations: lessons learned from marine reserves. Pages 280–302 in A. D. Falk, M. A. Palmer, and J. B. Zedler, editors. *Restoration ecology in context. Foundations of restoration ecology*. Island Press, Washington, D.C., USA.
- Osenberg, C. W., R. J. Schmitt, S. J. Holbrook, K. E. Abu-Saba, and A. R. Flegal. 1994. Detection of environmental impacts: natural variability, effect size, and power analysis. *Ecological Applications* 4:16–30.
- Osenberg, C. W., J. S. Shima, S. L. Miller, and A. C. Stier. 2011. Assessing effects of marine protected areas: confounding in space and possible solutions. Pages 143–167 in J. Claudet, editor. *Marine protected areas: a multidisciplinary approach*. Cambridge University Press, Cambridge, UK.
- R Core Team. 2017. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.
- Russ, G. R., and A. C. Alcala. 2004. Marine reserves: Long-term protection is required for full recovery of predatory fish populations. *Oecologia* 138:622–627.
- Russ, G. R., and A. C. Alcala. 2010. Decadal-scale rebuilding of predator biomass in Philippine marine reserves. *Oecologia* 163:1103–1106.
- Shears, N. T., R. V. Grace, N. R. Usmar, V. Kerr, and R. C. Babcock. 2006. Long-term trends in lobster populations in a partially protected vs. no-take Marine Park. *Biological Conservation* 132:222–231.
- Stewart-Oaten, A., and J. R. Bence. 2001. Temporal and spatial variation in environment impact assessment. *Ecological Monographs* 71:305–339.
- Stewart-Oaten, A., W. W. Murdoch, and K. R. Parker. 1986. Environmental impact assessment: “Pseudoreplication” in time? *Ecology* 67:929–940.

- Thiault, L., A. Collin, F. Chlous, S. Gelcich, and J. Claudet. 2017a. Combining participatory and socioeconomic approaches to map fishing effort in small-scale fisheries. *PLoS ONE* 12:e0176862.
- Thiault, L., L. Kernaléguen, C. W. Osenberg, and J. Claudet. 2017b. Progressive-change BACIPS: a flexible approach for environmental impact assessment. *Methods in Ecology and Evolution* 8: 288–296.
- Willis, T. J., R. B. Millar, R. C. Babcock, and N. Tolimieri. 2003. Burdens of evidence and the benefits of marine reserves: Putting Descartes before des horse? *Environmental Conservation* 30:S0376892903000092.
- Zupan, M., E. Fragkopoulou, J. Claudet, K. Erzini, B. Horta e Costa, and E. J. Gonçalves. 2018. Marine partially protected areas: drivers of ecological effectiveness. *Frontiers in Ecology and the Environment* 16:381–387.

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